

Fairchild and Hawthorne's sorites argument for permissivism

PHIL UA-78 Metaphysics

Prof. Cian Dorr

9th September 2026

Fairchild and Hawthorne

A Sorites sequence

[I]magine a continuum of physical transformations one could induce with a sanding machine applied to [a piece of] driftwood, ranging from minimal application to some edge to largescale transformation into a winerack of enduring beauty. No pair of cases seems to have any hope of marking an ontological difference between bringing a thing into existence and not bringing a thing into existence. The fact that one pair may mark the cutoff for some woolly ordinary language predicate like 'altered a lot by the sander' is not the kind of thing to have metaphysical import. (F&H 66)



A driftwood wine rack. Photographer unknown.

The structure of the argument

Let's consider a long series of possible worlds $w_1, \dots, w_n$, where at w_1 the sander is applied very vigorously to the driftwood in such a way that 50% of its mass is removed, while at w_n the sander barely grazes the wood.

P1: At w_1 , a new object is created when the sander is applied.

P2: There is no i such that at w_i a new object is created when the sander is applied, but at w_{i+1} , no new object is created when the sander is applied.

C: For every $i \leq n$, at w_i , a new object is created when the sander is applied.

► This argument is valid in classical logic.

How does this relate to the classic Sorites series?

A structurally similar argument: let $x_1, \dots, x_n$ be a long series of men, where each one has one more hair than the previous one, x_1 has no hairs, and x_n has a full head of hair.

P1: x_1 is bald.

P2: There is no i such that x_i is bald and x_{i+1} is not bald.

C: For every $i \leq n$, x_i is bald.

- The argument is valid (in classical logic); P1 is obvious, and C is absurd. So we have a reductio of P2.

How could removing one hair make someone bald?

It's hard to believe that applying the sander for one extra microsecond could make the difference between no new object being created and a new object being created. But it's *also* hard to believe that removing one hair could make someone bald!

Unless we start revising basic logical rules, we have to learn to live with the latter belief somehow. If we can do so, why not live with the first belief too?

Making one's peace with a last bald man

A common view: the word 'bald' doesn't just stand for a single property. It refers to (or maybe 'partially refers to') a whole bunch of properties. Each individual property has a cutoff, but they all have different cutoffs. So, as we add hairs one at a time, we never go straight from a man who has *all* the properties associated with the word 'bald' to a man who has *none* of those properties.

- This is the basic idea of what's called *supervaluationism* about vagueness, though some authors build various further commitments into their definition of that term.

How does this help? One thought: we find the idea of a last bald man initially absurd because we recognise that one hair can't make the difference between someone with all the properties referred to by 'bald' and someone with none of them, but we mistakenly assume that 'bald' only stands for a single property.

Supervaluationism and the sander Sorites

Could one say the same thing about the sander case? The idea is that 'Something came into existence' doesn't denote a single state of affairs, but a whole bunch of different ones, each of which ceases to obtain at a different point in the series.

But how could this work? What word is varying in its interpretation?

The most obvious candidate is the *quantifier*. But while it's easy to see how there could be a bunch of different meanings of the quantifier that are *restricted*, the idea of multiple *unrestricted* meanings for the quantifier seems incoherent.